

ETHAN MIKOLAYCIK

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EDUCATION

Texas A&M University

Bachelor of Science, Computer Engineering

College Station, TX

Aug 2025 – Expected 2029

Lake Travis High School

Austin, TX

High School Diploma

Aug 2021 – May 2025

- Published robotics tutorial series

EXPERIENCE

Software Lead Intern

Sep 2025 – Present

WHOOOP VEX U Robotics Team

College Station, TX

- Built mikGen, an open-source web-based path planning, simulation and code generation tool supporting mikLib, ReveilLib, and LemLib; adopted by 10+ teams during open beta
- Allowed rapid prototyping and development of autonomous routines and ReveilLib due to instant feedback
- Built WHOOPSim which allows drivers to control a simulated robot with a VEX controller
- Improved ReveilLib API by adding angle offsets, slew, named parameters, callbacks and automatic odometry tuning

PROJECTS

mikLib | C++, VEXcode

- Open-source library for VEXcode adding odometry, motion control, and an on-brain custom UI framework for real-time PID tuning and autonomous selection
- Implemented two-tracker IMU odometry, wall resetting, with turn, swing and drive to point, drive to pose, and follow path drive motions
- Implemented reusable PID class and motion chaining, which values can be individually passed into each drive motion; SD-card save/load for tuning constants and selected autonomous
- Documented with setup tutorials and Doxygen-style comments; actively maintained, and updated
- 450+ downloads on Github, helped numerous teams over Discord

Wall Reset Simulator | Javascript, HTML, CSS

- Built a browser-based simulator to validate a wall-reset localization algorithm, using virtual lasers to estimate robot pose from distances to field walls
- Implemented configurable front/left/right/rear lasers with sensor offsets, live visibility toggles, and field switching
- Deployed on GitHub Pages with documentation

Scene Maker | Java AWT/SWING

- Built a customizable scene maker; scene is built with background and particles, with velocity, size and amount parameters; snowflake and rain particles are pre-built
- Snowflakes are procedurally generated, and can be selected to rotate along the x, y and z axes
- Placed 1st in CS3 winter scene competition

Minigame Roulette | C# Unity

- Developed a Unity game that has a collection of randomized minigames, selected via a roulette wheel
- Implemented game selection logic and unified scoring across multiple minigame types
- Designed and programmed multiple interactive minigames with custom physics, movement, event handling and camera scripts
- Added UI for real-time score display and transitions that contain minigame controls

SKILLS

Programming Languages C++, Javascript, Typescript, Java, Python, C#

Technologies Git, GitHub, VEXcode, PROS, raylib, Unity, React, Tailwind, VS Code, Fusion 360

Interests Mountain biking, Lego, Origami